T3 group factors: UV

Active theory: SM + $S_1 + S_2 + F$. The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Field representations

run	S_1	S_2	F	α
T3 − B	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	$(1, 0)$	-1

T3 − C	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	$(3,0)$	-1
T3 − A	$(1,0)$	$(3,1)$	$(2,\frac{1}{2})$	0
T3 − D	$(3,-1)$	$(1,0)$	$(2,-\frac{1}{2})$	-2
T3 − E	$(3,0)$	$(3,1)$	$(2,\frac{1}{2})$	0

Gauge and Yukawa beta functions

$\beta^{(1)}_{g_1}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
g_1^3	$\frac{43}{6}$	$\frac{43}{6}$	$\frac{59}{6}$	$\frac{59}{6}$	$\frac{59}{6}$

$\beta^{(1)}_{g_2}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
g_2^3	$-\frac{17}{6}$	$\frac{7}{6}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{6}$

$\beta^{(1)}_{g_3}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
g_3^3	-7	-7	-7	-7	$-$

$\beta^{(1)}_{Y_u}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	$-$
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{1}{1}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\beta^{(1)}_{Y_d}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	$-$
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{1}{1}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\beta^{(1)}_{Y_e}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (

$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{1}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\beta_{y_1}^{(1)}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
$\mathrm{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	1	$\frac{3}{2}$	0	0	0
$T_\nu^{(1)}(y_1)_{ij}$	1	$\frac{3}{2}$	2	1	1
$(y_1(y_2)^\dagger y_2)_{ij}$	1	$\frac{1}{2}$	0	0	0
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{5}{4}$	1	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-$
$g_1^2(y_1)_{ij}$	$-\frac{3}{4}$	$-\frac{3}{4}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$

$\beta_{y_2}^{(1)}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E (
$\mathrm{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	1	$\frac{3}{2}$	0	0	0
$T_\nu^{(2)}(y_2)_{ij}$	1	$\frac{3}{2}$	1	2	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_2(y_1)^\dagger y_1)_{ij}$	1	$\frac{1}{2}$	0	0	0
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{3}{2}$	1	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-$
$g_1^2(y_2)_{ij}$	$-\frac{3}{4}$	$-\frac{3}{4}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-$

β_{M_F}

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$(y_1^T y_1^*)M_F$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	1	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{3}{4}$
$g_2^2 M_F$	0	-12	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 M_F$	0	0	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

$\beta_{m_1^2}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$m_1^2\, T_\nu^{(1)}$	2	3	4	2	2
$M_F^\dagger M_F\, T_\nu^{(1)}$	-16	-24	-8	-4	-4
$\lambda_{S_1}^{(1)}\, m_1^2$	6	6	4	8	8
$g_2^2\, m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	0	-12	-12
$g_1^2\, m_1^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	0	-6	0
$\lambda_{12}^{(1)}\, m_2^2$	4	4	6	2	6
$\lambda_{S_1}^{(A)}\, m_1^2$	0	0	0	4	4

$\lambda_{12}^{(\times)}\, m_2^2$	0	0	0	0	4

$\beta_{m_2^2}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$m_2^2\, T_\nu^{(2)}$	2	3	2	4	2
$M_F^\dagger M_F\, T_\nu^{(2)}$	−16	−24	−4	−8	−4
$\lambda_{S_2}^{(1)}\, m_2^2$	6	6	8	4	8
$g_2^2\, m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	−12	0	−12
$g_1^2\, m_2^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	−6	0	−6
$\lambda_{12}^{(1)}\, m_1^2$	4	4	2	6	6
$\lambda_{S_2}^{(A)}\, m_2^2$	0	0	4	0	4
$\lambda_{12}^{(\times)}\, m_1^2$	0	0	0	0	4

$\beta_{\lambda_{HS_1}^{(1)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$\lambda_{HS_1}^{(1)}\, T_\nu^{(1)}$	2	3	4	2	2
$\mathrm{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	−2	−3	−4	−2	−2
$\lvert\lambda_5\rvert^2$	6	6	12	4	4
$g_2^2\, \lambda_{HS_1}^{(1)}$	−9	−9	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\, \lambda_{HS_1}^{(1)}$	−3	−3	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_1\, \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(1)}$	6	6	4	8	8
$\lambda_{12}^{(1)}\, \lambda_{HS_2}^{(1)}$	4	4	6	2	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	6	6
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	0	3	0
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	2	2
$\lambda_{HS_1}^{(1)}\, T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)}\, \lambda_{S_1}^{(A)}$	0	0	0	4	4
$\lambda_{HS_2}^{(1)}\, \lambda_{12}^{(\times)}$	0	0	0	0	4

$\beta_{\lambda_{HS_2}^{(1)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$\lambda_{HS_2}^{(1)}\, T_\nu^{(2)}$	2	3	2	4	2
$\mathrm{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	−2	−3	−2	−4	−2
$\lvert\lambda_5\rvert^2$	6	6	4	12	4
$g_2^2\, \lambda_{HS_2}^{(1)}$	−9	−9	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$
$g_1^2\, \lambda_{HS_2}^{(1)}$	−3	−3	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_1\, \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(1)}$	6	6	8	4	8
$\lambda_{12}^{(1)}\, \lambda_{HS_1}^{(1)}$	4	4	2	6	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	6	0	6
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	3	0	3
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	2	0	2

$\lambda_{HS_2}^{(1)}\, T$	2	2	2	2	2
$\lambda_{HS_2}^{(1)}\, \lambda_{S_2}^{(A)}$	0	0	4	0	4
$\lambda_{HS_1}^{(1)}\, \lambda_{12}^{(\times)}$	0	0	0	0	4

$\beta_{\lambda_{12}^{(1)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$\mathrm{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	-2	$-\frac{9}{2}$	-4	-4	-2
$\lvert\lambda_5\rvert^2$	3	3	4	4	$\frac{4}{3}$
$g_2^2\, \lambda_{12}^{(1)}$	-9	-9	-12	-12	-24
$g_1^2\, \lambda_{12}^{(1)}$	-3	-3	-6	-6	-6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(1)}$	6	6	4	8	8
$\lambda_{12}^{(1)}\, \lambda_{S_2}^{(1)}$	6	6	8	4	8
$\lambda_{HS_1}^{(1)}\, \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	0	12
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	0	4
$\lambda_{12}^{(1)}\, T_\nu$	2	3	0	0	2
$\lambda_{12}^{(1)}\, T_\nu^{(1)}$	0	0	4	2	0
$\lambda_{12}^{(1)}\, T_\nu^{(2)}$	0	0	2	4	0
$\lambda_{12}^{(1)}\, \lambda_{S_2}^{(A)}$	0	0	4	0	4
$\lambda_{12}^{(1)}\, \lambda_{S_1}^{(A)}$	0	0	0	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	4
$\lambda_{12}^{(\times)}\, \lambda_{S_1}^{(A)}$	0	0	0	0	4
$\lambda_{12}^{(\times)}\, \lambda_{S_2}^{(A)}$	0	0	0	0	4

β_{λ_5}

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$\lambda_5\, \lambda_1$	2	2	2	2	2
$\lambda_5\, \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5\, \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5\, \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2\, \lambda_5$	-9	-9	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$
$g_1^2\, \lambda_5$	-3	-3	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lambda_5\, \lambda_{HS_1}^{(A)}$	-2	-2	0	-4	-2
$\lambda_5\, \lambda_{HS_2}^{(A)}$	2	2	4	0	2
$\lambda_5\, \lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0	0	2
$\lambda_5\, T$	2	2	2	2	2
$\lambda_5\, T_\nu$	1	$\frac{3}{2}$	0	0	1
$\lambda_5\, T_\nu^{(1)}$	0	0	2	1	0
$\lambda_5\, T_\nu^{(2)}$	0	0	1	2	0
$\lambda_5\, \lambda_{12}^{(\times)}$	0	0	0	0	-2

Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$g_1^2 g_2^2$	-6	-6	0	-12	0

$\beta_{\lambda_{HS_2}^{(A)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$g_1^2 g_2^2$	6	6	12	0	12

$\beta_{\lambda_{12}^{(A)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
$g_1^2 g_2^2$	-6	-6	0	0	0

$\beta_{\lambda_{12}^{(\times)}}$

$\mathcal{S}$	T3 − B ($\alpha=-1$)	T3 − C ($\alpha=-1$)	T3 − A ($\alpha=0$)	T3 − D ($\alpha=-2$)	T3 − E ($\alpha=0$)
g_2^4	0	0	0	0	6
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	2
$\lambda_{12}^{(1)}\,\lambda_{12}^{(\times)}$	0	0	0	0	8
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	10
$\lambda_{12}^{(\times)}\,\lambda_{S_1}^{(A)}$	0	0	0	0	-2
$\lambda_{12}^{(\times)}\,\lambda_{S_2}^{(A)}$	0	0	0	0	-2

First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether F, S_1, S_2, or a degenerate set is integrated out first.